

Intro

Hello and welcome to the Cargoful tech interview!

We have prepared one exercise to assess programming and software engineering knowledge.

We expect to receive the results via mail or any other sharing service you might like (e.g. Github, Google Drive, etc..). We don't have strict expectations on the deliverables, so feel free to choose the medium you prefer, it's up to you!

Exercise

The main goal of the exercise is to create a web page in a modern SaaS web application dedicated to monitoring the execution of scheduled automations (Zapier style).

We ask you to develop both Front and Back End that let the user add new automations and interact with them.

Front end

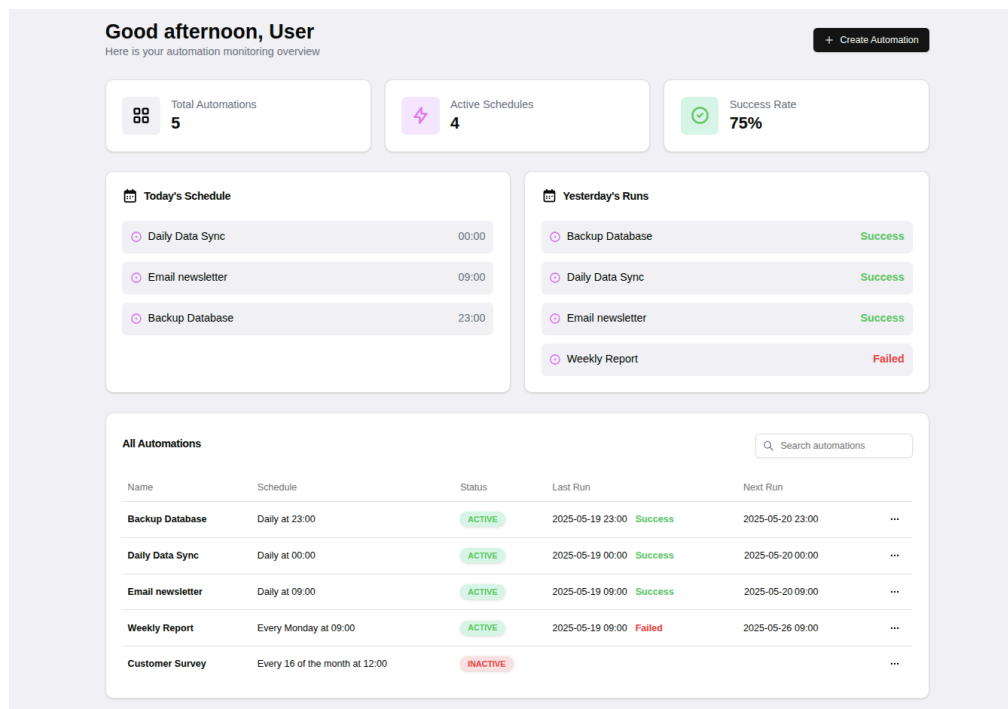
The page should include the following sections:

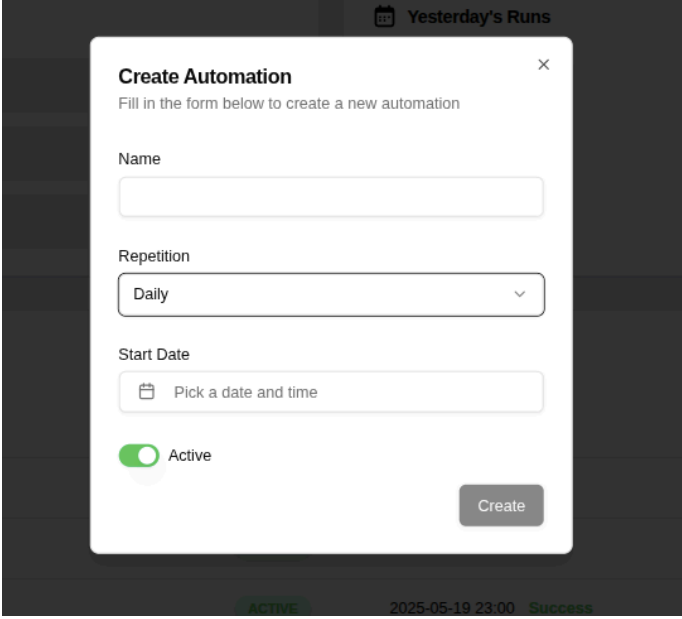
- welcome to the user and create automation button,
- summary KPIs on:
 - total number of automation,
 - active schedule automations,
 - a success rate of the automations (count on “last run”)
- a list of the automations scheduled for today,
- a list of the automations actually ran yesterday,
- a grid with all the automation in table format.

User Interaction:

1. Clicking on the active schedules box will filter the table showing only the active ones;
2. Clicking on the success rate box will filter the table showing only the successful automations
3. The search automation box searches on all the displayed fields e.g.: searching for “09:00” will show both the rows with schedule at 09:00 and rows with last run at 09:00.

Here’s a mockup of the page. Please develop the page in React. Feel free to use any state management packages like Recoil/Jotai/Redux and components libraries as Material UI.





Yesterday's Runs

Create Automation

Fill in the form below to create a new automation

Name

Repetition

Daily

Start Date

Pick a date and time

Active

Create

2025-05-19 23:00 - Success

Backend

We'd like you to implement a simple backend that supports **CRUD operations** for an **automation entity**. You're free to structure and implement the API as you prefer, as long as it fulfills the following requirements:

- **Required operations:**
 - The **frontend currently only uses Create and Read** endpoints.
(Bonus: if you feel like going the extra mile, feel free to wire up Update/Delete on the frontend too 😊)
 - Implement a way to keep track of automation execution (when an automation runs)
- **Database:**

Use **SQLite** for the database so you can easily share the DB file with us.

You're free to design the **database schema** as you prefer.
- **Automation data:**

Since the automations won't actually run, you can manually populate or edit the database to include **meaningful placeholder values** (e.g., for fields like `last_run`).
- **Framework:**

Please use **Django** for the backend implementation.
- **Authentication:**

No login or authentication is needed—keep it simple and focused on the core logic.

Let us know if you have any questions or need clarification at any stage!